

CSC1109 – Data (Analytics) at Speed and Scale

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Apache Hadoop (briefly)

Open source software framework consisting of

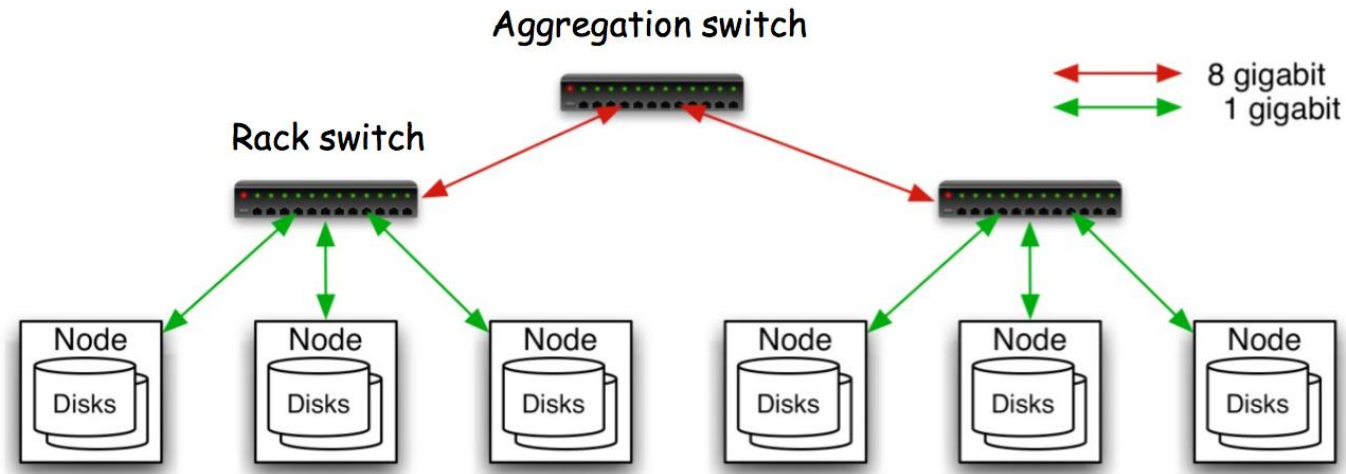
1. Distributed Storage
 - Hadoop Distributed File System
2. Parallel Processing model
 - MapReduce
3. Resource Manager
 - TaskTracker/Hadoop YARN

Typical Hadoop Cluster



Cluster of machines running Hadoop @Yahoo!

Typical Hadoop Cluster



- 40 nodes/rack, 1000-4000 nodes in cluster
- 1Gbps bandwidth within rack, 8 Gbps out of rack
- Node specs:
 - 8x2GHz cores, 8GB RAM, 4x1 TB disks



- A Mapreduce **Job** splits input data (usually files) into chunks
 - InputSplit & RecordReader
- The Chunks are sent to **map** tasks via the TaskTracker / YARN
 - Optional Combiner task
- The output of the map tasks is **sorted** and **routed** to **reduce** tasks
 - Shuffle & Sort happen simultaneously (retrieve and group intermediates)
- The **output** of the **reduce** tasks is written to files
 - Output goes to a file (typically)

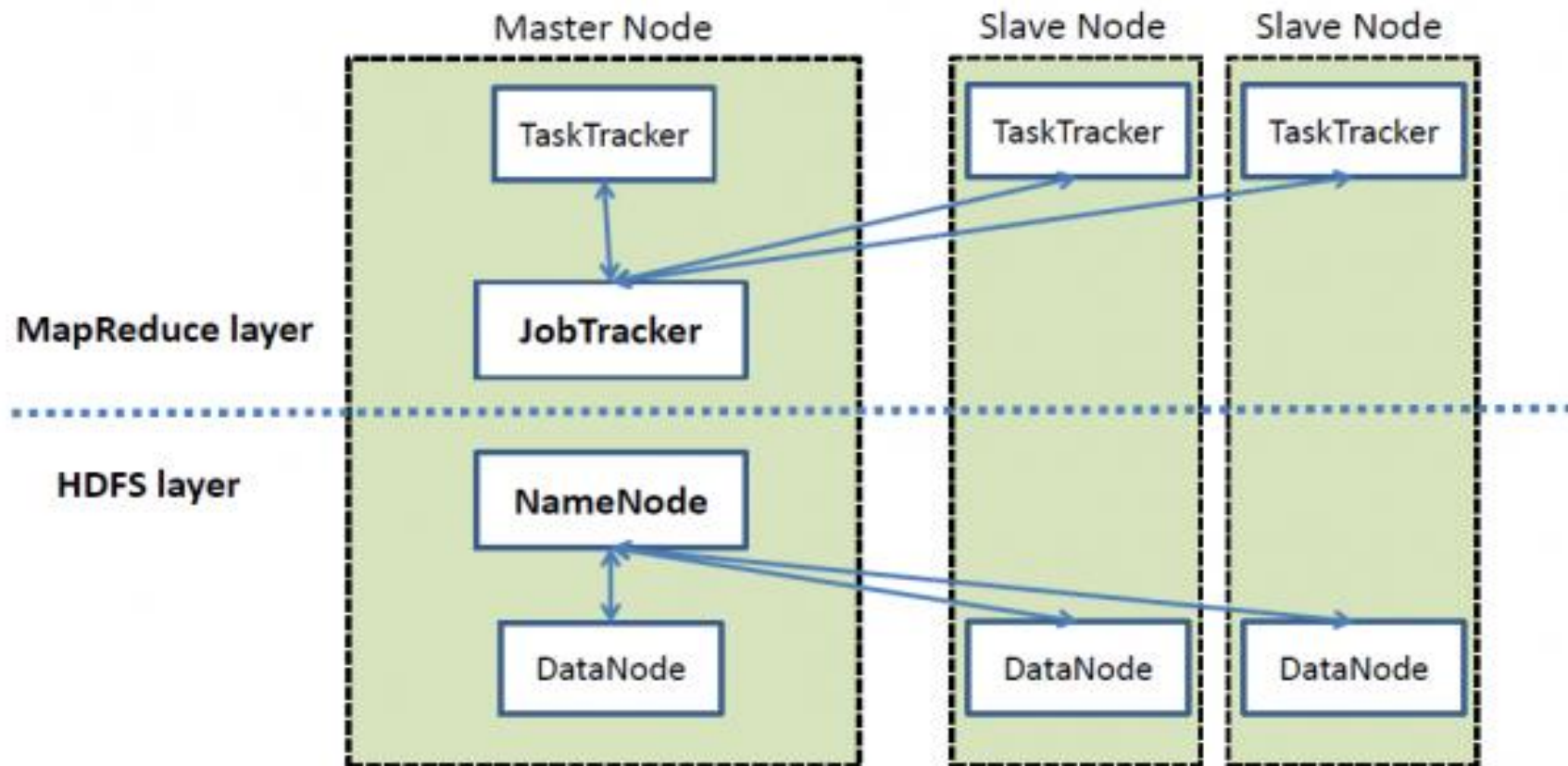
Hadoop...

- handles scheduling (which work tasks are executed when)
- assigns workers to map- and reduce-jobs
- distributes data among several nodes in the cluster
- executes jobs, ideally locally in proximity to the respective data
- synchronisation
- gathering, shuffling, grouping... of intermediate data
- error handling, fault tolerance

HDFS – Hadoop Distributed File System

- NameNode
 - Dedicated server for metadata
 - Holds the directory of where all the files are kept on the cluster
 - Does not hold the content of the files itself
- DataNode
 - Servers storing application data
 - Register with the NameNode
 - Talk directly to the TaskTracker and Client applications, once found via the NameNode
- MR layer
 - TaskTracker node accepts tasks from the JobTracker and routes them to the nearest open slot (ideally on the DataNode with that data)
 - This is replaced from Hadoop 2.0 with YARN

High Level Architecture of Hadoop



Hadoop Locality

- Allocate work (jobs) on the cluster node where the respective data is
 - Move computation to the data
 - Reduce possible network traffic (which also depends on distance between machines)
- Priority is given to machines in same cluster
- Good knowledge of hardware and network technology crucial for setting up large Hadoop clusters

Hadoop 2.0

Applications Run Natively **IN** Hadoop

BATCH
(MapReduce)

INTERACTIVE
(Tez)

ONLINE
(HBase)

STREAMING
(Storm, S4,...)

GRAPH
(Giraph)

IN-MEMORY
(Spark)

HPC MPI
(OpenMPI)

OTHER
(Search)
(Weave...)

YARN (Cluster Resource Management)

HDFS2 (Redundant, Reliable Storage)

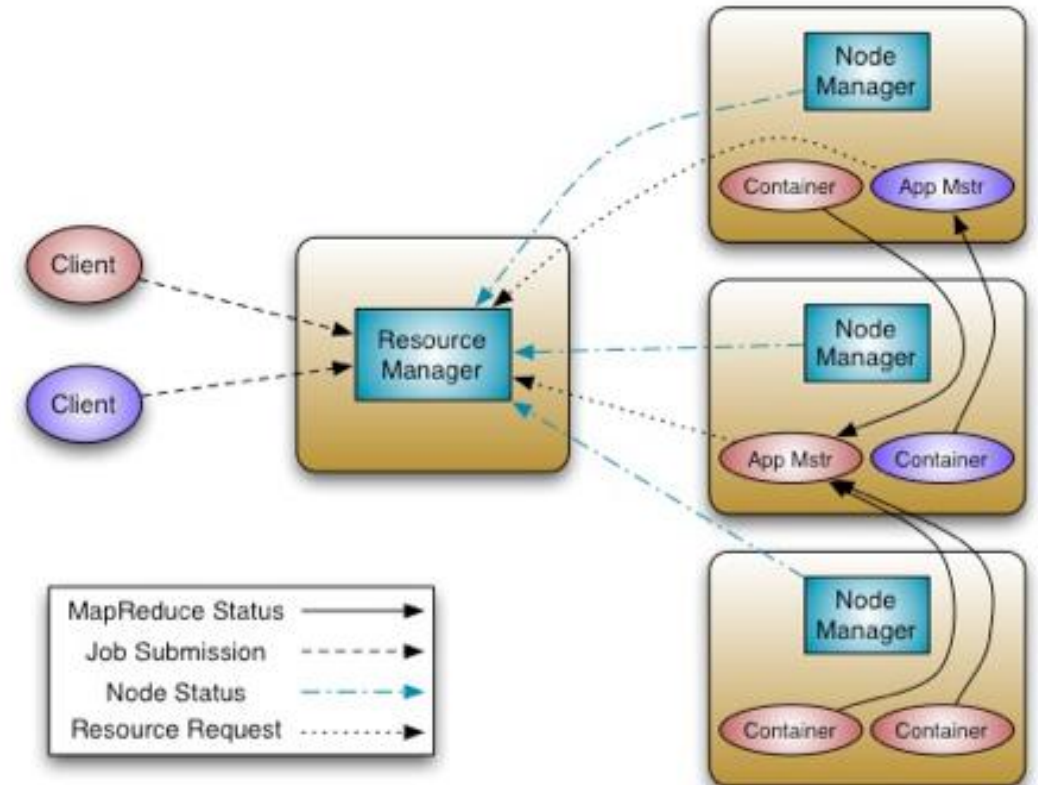


About YARN

- YARN=Yet Another Resource Negotiator
- Manages and monitors workloads
- Organizes applications into DAGs of jobs, under common resource management
- Single platform for multiple data processing engines:, streaming data and conventional data (SQL) can be hosted and processed under the same resource management (provided by YARN))
- Provides a multi-tenant environment (multiple groups of users can access the same data and same processing engines)
- Also implements security, ensures high availability of cluster...

YARN (or Mapreduce 2.0)

- Split the JobTracker into two roles:
 - Global ResourceManager
 - Global Authority
 - Pluggable Scheduler
 - Per-application ApplicationMaster
 - Application-specific negotiator for resources
- Add a Per-node NodeManager
- Add a Per-Application/per-Node Container



End of Content